Zeyu Hu

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Ph.D. candidate in Statistics with hands-on experience in applied biostatistics, including analysis of clinical trials, longitudinal aging cohorts, infectious disease surveillance, functional brain networks, and large-scale genomics. Strong foundation in Bayesian modeling, MCMC computation, complex network analysis and longitudinal modeling, with experience carrying novel statistical methods from derivation through simulation validation to open-source R packages, supported by reproducible R/Python workflows (Linux/Git/HPC) to deliver interpretable results, publication-quality figures and recommendations for research and clinical teams.

EDUCATION

Doctor of Philosophy: Statistics — Sep 2023 – Dec 2026 (expected)
University of Connecticut - Storrs, CT

Master of Arts: Statistics — Sep 2020 – May 2022
Boston University - Boston, MA

Bachelor of Science: Applied Statistics — Sep 2016 – May 2020
Sun Yat-Sen University - Guangzhou, China

SKILLS

Statistical Methods: Bayesian modeling and computation (MCMC), network analysis, community detection, mixed-effects and longitudinal models, survival analysis, spatiotemporal and time-series modeling, causal inference.
Clinical Trial Design & Evaluation: Bayesian borrowing, robust mixture prior, effective sample size, operating characteristics, power and type I error assessment, continuous, count and time-to-event endpoints.
Programming & Workflows: R, SAS, Python, SQL, Git, HPC, Linux, Bash, Docker.
Application Domains: pediatric clinical trials, functional neuroimaging and Alzheimer's disease, genomics, aging and gerontology, infectious disease epidemiology.

INDUSTRY EXPERIENCE

Biostatistics Intern — May 2026 – Aug 2026
Insmed Incorporated - Bridgewater, NJ

- Developed and validated Bayesian workflows for pediatric clinical trials, using robust mixture priors to incorporate adult evidence for continuous, count and time-to-event endpoints.
- Designed simulation studies to quantify operating characteristics, including type I error, power, and bias, along with effective sample size across borrowing weights and varying degrees of prior-data conflict.
- Implemented a modular, end-to-end, version-controlled R framework spanning data preparation, simulation, and reporting, and integrated it into interactive R Shiny applications for user-directed evaluation of information-borrowing scenarios, with an extensible architecture designed to support Bayesian projects with analogous model structures.
- Developed SAS programs to support future application of the framework to real clinical trial data, enabling assessment of the robustness and practical implications of model and prior specifications and generation of decision-ready visualizations and technical recommendations for statistical and clinical stakeholders.

Big Data Engineering Intern, Research and Development Center — May 2019 – Aug 2019
Agricultural Bank of China - Guangzhou, China

- Helped build a data classification and search platform for thousands of financial tables from Guangdong branches.
- Applied K-means clustering and spectral clustering algorithms to merge tables, making it easier for staff to find information from massive data and significantly reduced storage requirements.

RESEARCH EXPERIENCE

Graduate Research Assistant, Department of Statistics — Sep 2023 – Present
University of Connecticut – Storrs, CT

- Developed **RCDC**, a Bayesian MCMC community detection method for noisy networks with covariates; applied it to ADNI and Human Connectome Project fMRI data, capturing Alzheimer's-related and individual-level variation.

- Developed **C4**, an adaptive spectral clustering algorithm combining weighted networks and node covariates through data-driven weighting; revealed eight geographically coherent communities in a North American airport network.
- Developed **BADGERS**, a graph-guided Bayesian variable-selection model with measurement-error correction for high-dimensional data, achieving higher prediction accuracy than existing methods and identifying relevant genes in ADNI and COPD studies.
- Developing a graph-guided Bayesian factor model for mixed-type, high-dimensional multi-study data, with a planned ADNI application to identify shared, partially shared, and study-specific genomic patterns.

Graduate Research Assistant, Center on Aging Jan 2026 – Present
UConn Health – Farmington, CT

- Quantify longitudinal associations among appetite loss, inflammatory biomarkers, and incident physical disability in community-dwelling older adults using mixed-effects models and Fine–Gray competing risks regression.
- Model longitudinal associations between metabolic syndrome and urinary incontinence among women, accounting for demographic, behavioral, and clinical factors to characterize urinary symptom patterns over time.

Graduate Research Assistant, Department of Public Health Sciences Oct 2023 – Jan 2026
UConn Health – Farmington, CT

- Quantified dynamic interactions among COVID-19 incidence, population mobility, and social-distancing policies using vector autoregression, characterizing the timing and persistence of policy effects.
- Modeled time to recovery from long COVID using accelerated failure time models and delivered analyses, figures, and manuscript sections for an international team of public health investigators.
- Modeled how consumer spending tracked mobility across 25 districts and 8 sectors with credit-card transaction data and fixed-effects panel regression, revealing sector-specific decoupling during pandemic recovery.
- Constructed district-level mobility networks across Seoul to identify how urban connectivity reorganized during the pandemic.

SELECTED PUBLICATIONS

- **Hu, Z.**, Li, W., Yan, J., & Zhang, P. (2026). Covariate connectivity combined clustering for weighted networks. *Physica A: Statistical Mechanics and its Applications, 704*, 132177.
- **Hu, Z.**, Yu, M., Jang, Y., Shrestha, S., Jung, J., & Jo, Y. (2026). Complex network analysis of mobility dynamics in Seoul during the COVID-19 pandemic 2020–2022. *Scientific Reports, 16*, 4345.
- **Hu, Z.**, Chang, W., Jang, Y., Jo, Y., & Jung, J. (2025). Dynamic interactions of COVID-19 incidences, mobility, and social distancing policies in Seoul: A VAR model approach. *Scientific Reports, 15*, 24288.
- Jo, Y., **Hu, Z.**, Chang, W., Jang, Y., & Jung, J. (In press). Changes in mobility and expenditure during the COVID-19 pandemic and social distancing in Seoul Metropolitan City. *BMC Public Health*.
- Jo, Y., **Hu, Z.**, Joo, H., Jung, J., & Lee, J. (2026). Time to recovery from long COVID: A longitudinal analysis of symptom duration and risk factors using accelerated failure time models. *International Journal of Infectious Diseases*, 108963.
- **Hu, Z.**, Xu, F. H., Kundu, S., Zhao, Y., Shen, L., Yan, J., & Li, W. (2026). Robust community detection for noisy networks with covariates: Application to functional brain networks. arXiv:2609.05654. Under review at *Biometrics*.

OPEN-SOURCE SOFTWARE

- c4 (R package): Covariate Connectivity Combined Clustering.
- rcdc (R package): Robust Community Detection for noisy networks with Covariates.

PRESENTATIONS

- **Data Blitz Talk & Poster**, Robust Community Detection for Noisy Networks with Covariates: Application to Functional Brain Networks, 29th Annual Neuroscience at Storrs, University of Connecticut, Oct 2026.
- **Poster**, Longitudinal Study of Appetite Loss and Inflammation in Older Adults: The Health, Aging, and Body Composition Study, Gerontological Society of America Annual Scientific Meeting, Nov 2026 (accepted).
- **Invited Talk,** The Impact of Covid-19 and social distancing policy on mobility and expenditure in Seoul Metropolitan City, COVID-19 Data and Disease Modeling International Symposium, Incheon, South Korea, May 2024.

PROFESSIONAL SERVICE

- **Peer Reviewer**: *The Annals of Applied Statistics.*
- **Memberships**: American Statistical Association (ASA), Institute of Mathematical Statistics (IMS), New England Statistical Society (NESS), and International Chinese Statistical Association (ICSA).